What the Health Navigator

A What-if Tool for Large Language Models

Elif Özsoy[1], Zhijing Liu[1], Benjamin Dettling[1], Alara Egi[1]
[1] D-INFK, ETH Zurich

1 Introduction

This project provides an interactive tool to explore how large language models (LLMs) like GPT generate disease predictions based on user-provided inputs.

The tool is designed to enhance transparency and enable explainability in AI-driven medical diagnosis systems.

2 Method Overview

Goals and Approach

- Enable users to interactively explore LLM-generated predictions by manipulating inputs
- Use a bar chart to visualize the probability distribution of diseases and highlight changes in predictions
- Provide provenance tracking with history features for user-submission comparisons

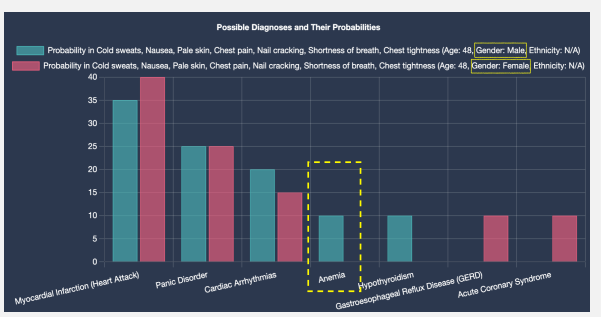
Implementation Highlights

- Backend: Utilized LLM for predictions based on inputs (symptoms, age, gender, ethnicity)
- Frontend: Designed an accessible and responsive interface using React and Chart.js for visualization
- Visualization: Dynamic bar charts showing changes in predictions for easy comparison

3 Usage Scenario

Dr. Emma, a trainee physician, has been following discussions about AI's growing role in medical diagnostics and its potential to revolutionize healthcare, even raising concerns about replacing human expertise. To evaluate the credibility of such claims, she devises a hypothetical case involving anemia:

Statistically, women have a higher likelihood of being anemic compared to men. However, the model produces an unexpected result: a patient with anemia-related symptoms is assigned a 5% probability of anemia when specified as male. Changing the input to female reduces the probability to 0%, contradicting established medical data.

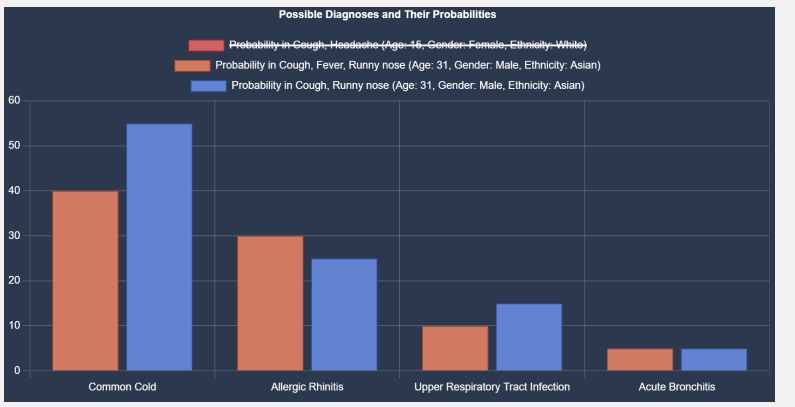


Screenshot from our website

4 Results and Discussion

This chart compares diagnosis probabilities generated by a large language model using different inputs (symptoms, age, gender, ethnicity). Users can observe how each input affects the results. For example, adding "Fever" to "Cough, Runny Nose" changes the probabilities significantly. The chart is interactive, allowing users to toggle specific datasets by clicking on the legend, making it easier to focus on individual comparisons and their impacts.

This interactive visual representation emphasizes the dynamic nature of medical diagnostic modeling and aids in exploring the impact of individual variables in clinical predictions.



Screenshot from our website

5 Conclusion

This project demonstrates the potential of an interactive tool to explore and analyze predictions generated by large language models (LLMs) in medical diagnosis. By allowing users to manipulate input parameters and visualize probability distributions, the tool enhances transparency, provides insights into the impact of individual attributes on predictions, and allows users to find counterfactual examples to identify and evaluate the model's limitations. The system promotes explainability in AI-driven healthcare applications, helping users understand and trust model outputs.
Future improvements could focus on further highlighting the differences in attributes, adding the option of changing the underlying model, and refining the visualizations.
This marks a step forward in understanding how various attributes contribute towards the output of an LLM.

Related Work

1. F. Cheng et al, Interactive Analysis of LLMS using Meaningful Counterfactuals, 2024.
2. Gomez and Holter, et al, Visual Counterfactual Explanations for Machine Learning Models, 2020.
3. What-If Tool, Google PAIR: https://pair-code.github.io/what-if-tool/

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